

Frederick Houses Part 3

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Getting Set Up

In addition to our usual `tidyverse` package, we load

- `scales` for making nice axis labels with variables measured in dollars. Use `+ scale_x_continuous(labels = label_dollar())`.
- `broom` for using models to “augment” data sets with fitted values and residuals.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(broom)
```

The data

We will again study the dataset that includes records of sales of residential properties in Frederick County from 2019.

The next chunk loads the data and adjusts two of the variables to make them a bit easier to graph and summarize.

```
frederick_2019 <- read_csv("data/frederick_2019.csv") %>%
  mutate(basement = factor(basement, levels = c("NO", "YES", "FINISH")),
         heat = fct_infreq(heat))

## Rows: 3284 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (2): basement, heat
## dbl (6): price, land_area, building_area, full_baths, half_baths, condition
## lgl (1): has_fireplace
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Brief codebook

The cases in the data set are the sales of residential properties in Frederick County in 2019. If a property was sold multiple times in 2019, only the last sale is recorded. A relatively small number of sales (on the order of 100) were removed from the dataset since their records were incomplete.

Variables

- **price**: the sales price of the property, in USD.
- **land_area**: the area of the land associated with the property, measured in acres
- **building_area**: the area of the main building on the property, not including “below grade” space such as a basement, in square feet.
- **heat**: the type of heat for the main building
- **full_bath**: the number of full baths (bathrooms with bathing facilities) in the main building
- **half_bath**: the number of half baths (bathrooms without bathing facilities) in the main building
- **condition**: condition of the main building on an 8-point scale, where 8 is “excellent”
- **has_fireplace**: whether the main building has at least one fireplace

Examine the dataset in the usual ways and familiarize yourself with the variables and their types.

```
head(frederick_2019, 10)
```

```
## # A tibble: 10 x 9
##   price land_area building_area basement heat full_baths half_baths condition
##   <dbl>   <dbl>       <dbl> <fct>   <fct>   <dbl>   <dbl>   <dbl>
## 1  50000    0.74        1600 YES     SPACE     1         0         3
## 2  75000    0.25         992 YES     SPACE     0         0         2
## 3 300000    0.43        1080 FINISH  ELEC     2         1         4
## 4  40000    0.12         880 NO      SPACE     1         0         2
## 5 332000    0.22        2080 YES     HT P~    2         1         4
## 6 318500    2.42        1298 YES     HT P~    2         0         4
## 7 265000    1.39        1092 YES     HT P~    2         0         4
## 8 425000    1.92        2136 YES     HT P~    3         0         4
## 9 212100    0.116       1196 YES     HT P~    1         0         3
## 10 330000    1.09        1804 YES     HT A~    2         0         4
## # i 1 more variable: has_fireplace <lgl>
```

```
glimpse(frederick_2019)
```

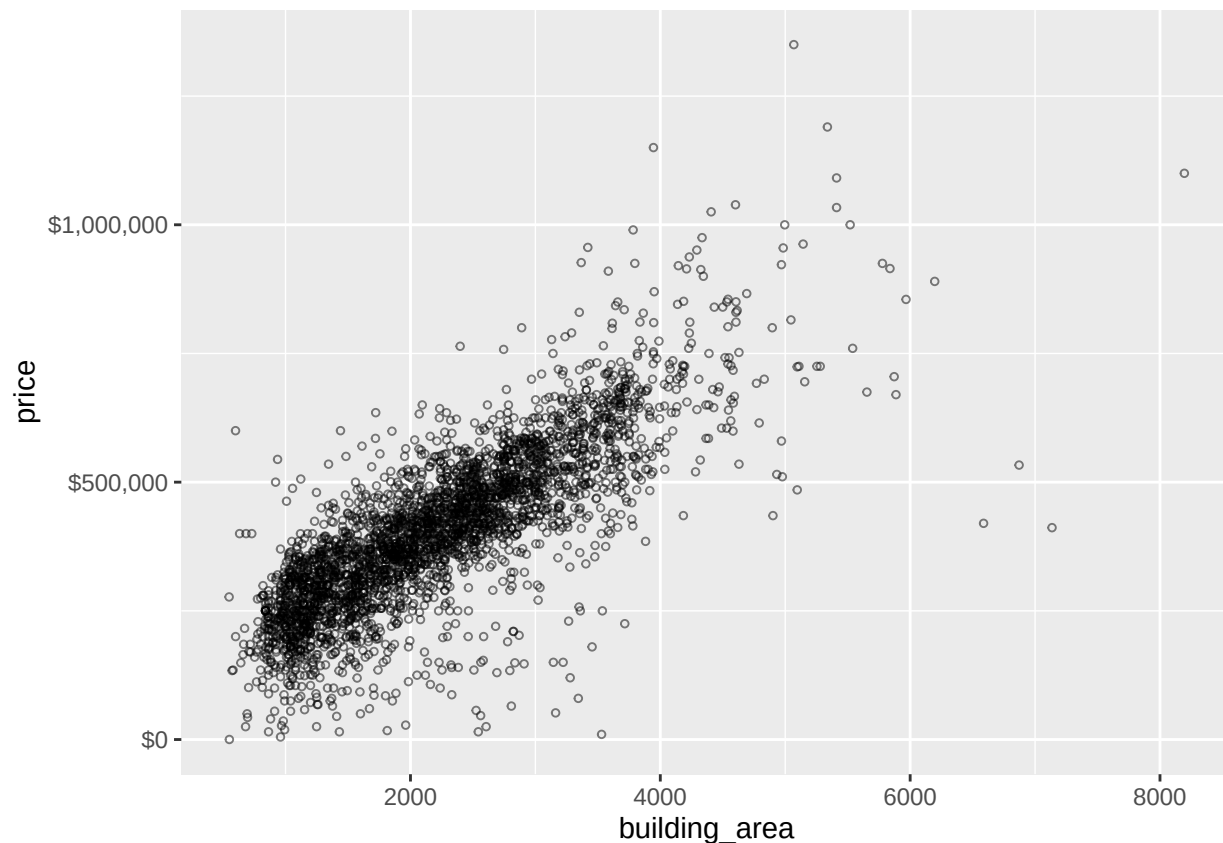
```
## Rows: 3,284
## Columns: 9
## $ price      <dbl> 50000, 75000, 300000, 40000, 332000, 318500, 265000, 425~
```

```
## $ land_area      <dbl> 0.7400, 0.2500, 0.4300, 0.1200, 0.2200, 2.4200, 1.3900, ~
## $ building_area <dbl> 1600, 992, 1080, 880, 2080, 1298, 1092, 2136, 1196, 1804~
## $ basement      <fct> YES, YES, FINISH, NO, YES, YES, YES, YES, YES, YES, YES,~
## $ heat          <fct> SPACE, SPACE, ELEC, SPACE, HT PMP, HT PMP, HT PMP, HT PM~
## $ full_baths     <dbl> 1, 0, 2, 1, 2, 2, 2, 3, 1, 2, 2, 1, 1, 2, 1, 1, 2, 3, 1,~
## $ half_baths     <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 1,~
## $ condition      <dbl> 3, 2, 4, 2, 4, 4, 4, 4, 3, 4, 4, 3, 4, 3, 4, 4, 4, 5, 5,~
## $ has_fireplace <lgl> TRUE, FALSE, TRUE, FALSE, FALSE, TRUE, TRUE, TRUE, FALSE~
```

Here, we are interested in predicting the sales price of the houses again! This time, we will use two predictor variables: both the square footage of the building area (numerical) and whether the house has at least one fireplace or not (categorical).

1. To get started, run the code below to see the (now) familiar graph of price versus building area.

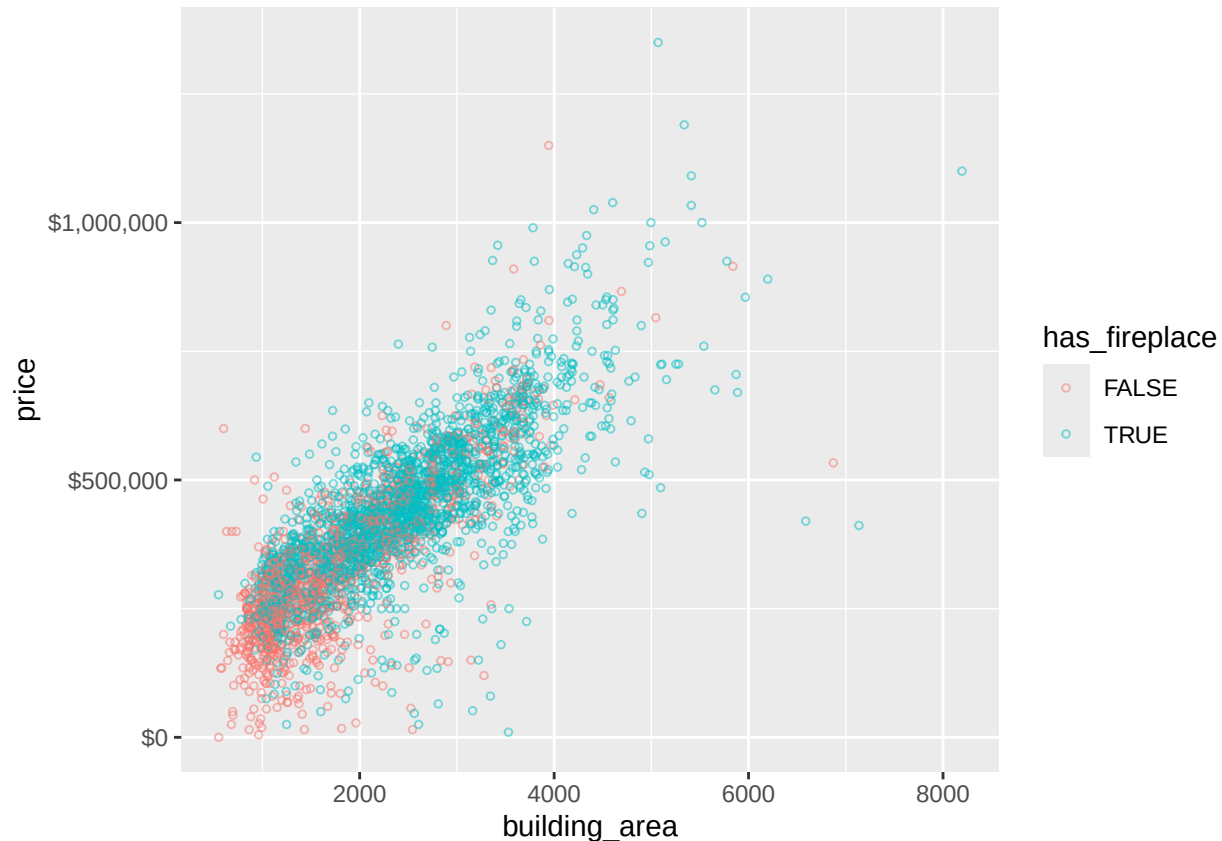
```
ggplot(data = frederick_2019,
       mapping = aes(x = building_area,
                     y = price)) +
  geom_jitter(shape = 1, size = 1, alpha = 0.5) +
  scale_y_continuous(labels = label_dollar())
```



Whenever we predict values of a numerical variable using two explanatory variables where one is numerical and the other is categorical, we can visualize the relationship by making a scatterplot of the numerical response versus the numerical predictor and then coloring the points by the categorical variable.

2. Copy and paste the code for the scatterplot of price vs building_area below. Then modify the scatterplot to display the variable has_fireplace as well by coloring the points in the scatterplot according to the variable has_fireplace. To do this, add a color = has_fireplace argument to the ggplot mapping aesthetic.

```
ggplot(data = frederick_2019,
       mapping = aes(x = building_area,
                     y = price,
                     color = has_fireplace)) +
  geom_jitter(shape = 1, size = 1, alpha = 0.5) +
  scale_y_continuous(labels = label_dollar())
```



3. Next, we will use lm to make a multiple linear regression model predicting price using the *two* predictor variables: building_area and has_fireplace. To do this, put building_area + has_fireplace where the explanatory (predictor) variable usually goes in the lm function. Notice there is a plus sign between the two predictor variables.

```
fit <- lm(price ~ building_area + has_fireplace, data = frederick_2019)
summary(fit)
```

```
##
## Call:
## lm(formula = price ~ building_area + has_fireplace, data = frederick_2019)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -645118  -45593    3779   53121  562917
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   96610.841    4309.990   22.416 < 2e-16 ***
## building_area    130.448      1.846   70.646 < 2e-16 ***
## has_fireplaceTRUE 29232.836    3623.098    8.068 9.89e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 93780 on 3281 degrees of freedom
## Multiple R-squared:  0.6505, Adjusted R-squared:  0.6503
## F-statistic: 3053 on 2 and 3281 DF,  p-value: < 2.2e-16
```

4. Print the model summary output if you have not already done so above.

```
summary(fit)

##
## Call:
## lm(formula = price ~ building_area + has_fireplace, data = frederick_2019)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -645118  -45593    3779   53121  562917
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   96610.841    4309.990   22.416 < 2e-16 ***
## building_area    130.448      1.846   70.646 < 2e-16 ***
## has_fireplaceTRUE 29232.836    3623.098    8.068 9.89e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 93780 on 3281 degrees of freedom
## Multiple R-squared:  0.6505, Adjusted R-squared:  0.6503
## F-statistic: 3053 on 2 and 3281 DF,  p-value: < 2.2e-16
```

5. Use the model summary to write what you think the equation for the model will be.

$\text{price_hat} = 96610.841 + 130.448(x_1) + 29232.836(x_2)$ where x_1 is `building_area`, and x_2 is `has_fireplace`

Creating a graph of this model isn't quite as easy as it was when we had just a single numerical predictor variable.

Before going ahead, view the `frederick_2019` dataframe to remind yourself of the variable names. We are about to modify the dataframe so it will be good for us to notice what changes.

6. Augment the data frame with the details for the model. To do this, we use the `augment()` function from the `broom` package, as we have done before. Run the code chunk.

```
frederick_2019 <- fit %>%
  augment(frederick_2019)
```

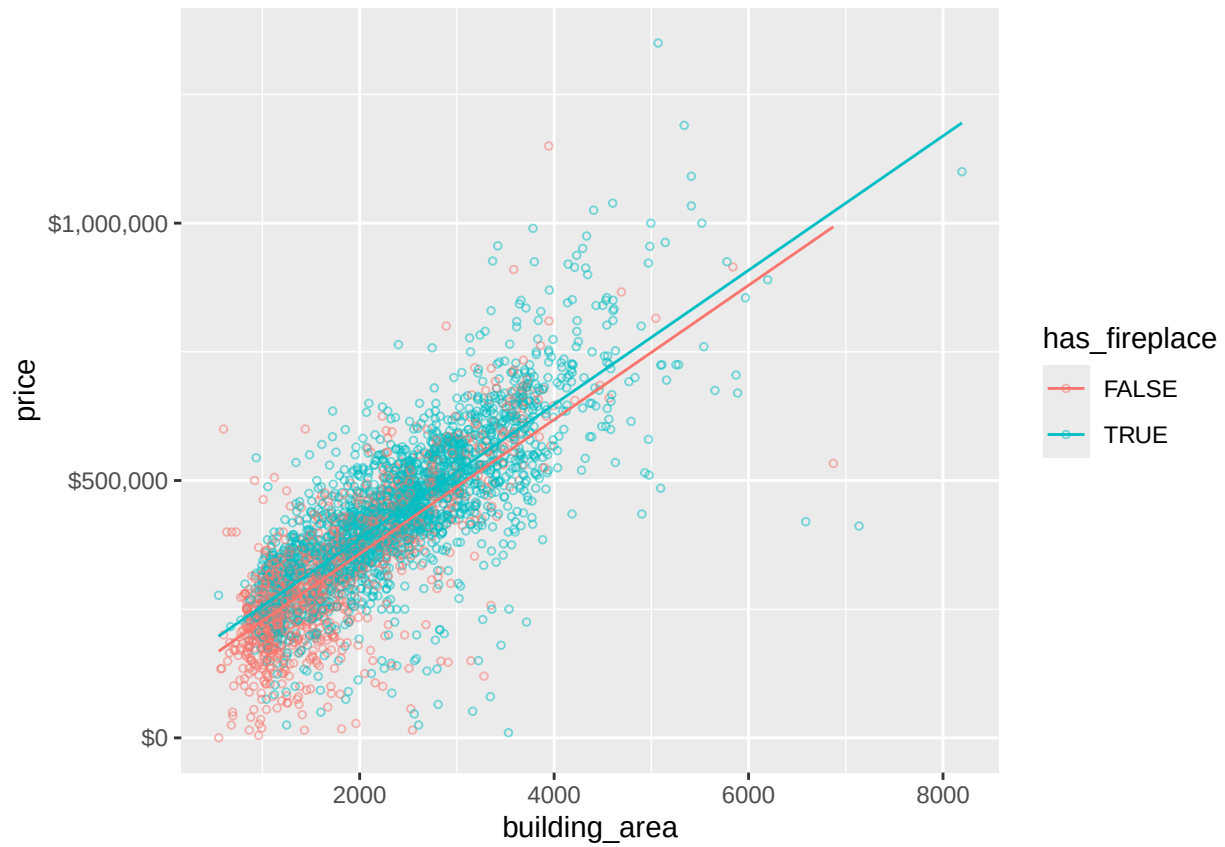
7. View the dataframe again to see that a number of new columns have been added. Notice in particular that the fitted values for the model are included as the variable `.fitted`.

```
glimpse(frederick_2019)
```

```
## Rows: 3,284
## Columns: 15
## $ price      <dbl> 50000, 75000, 300000, 40000, 332000, 318500, 265000, 425~
## $ land_area  <dbl> 0.7400, 0.2500, 0.4300, 0.1200, 0.2200, 2.4200, 1.3900, ~
## $ building_area <dbl> 1600, 992, 1080, 880, 2080, 1298, 1092, 2136, 1196, 1804~
## $ basement   <fct> YES, YES, FINISH, NO, YES, YES, YES, YES, YES, YES, YES,~
## $ heat        <fct> SPACE, SPACE, ELEC, SPACE, HT PMP, HT PMP, HT PMP, HT PM~
## $ full_baths  <dbl> 1, 0, 2, 1, 2, 2, 2, 3, 1, 2, 2, 1, 1, 2, 1, 1, 2, 3, 1,~
## $ half_baths  <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 1,~
## $ condition  <dbl> 3, 2, 4, 2, 4, 4, 4, 4, 3, 4, 4, 3, 4, 3, 4, 4, 4, 5, 5,~
## $ has_fireplace <lgl> TRUE, FALSE, TRUE, FALSE, FALSE, TRUE, TRUE, TRUE, FALSE~
## $ .fitted     <dbl> 334559.9, 226014.9, 266727.1, 211404.8, 367941.9, 295164~
## $ .resid      <dbl> -284559.8825, -151014.8886, 33272.8845, -171404.7542, -3~
## $ .hat        <dbl> 0.0007540411, 0.0011088614, 0.0012022788, 0.0011839382, ~
## $ .sigma      <dbl> 93660.65, 93755.34, 93790.65, 93744.63, 93790.35, 93791.~
## $ .cooksd     <dbl> 2.317784e-03, 9.606293e-04, 5.057156e-05, 1.321536e-03, ~
## $ .std.resid  <dbl> -3.035538750, -1.611235400, 0.355017687, -1.828851376, --
```

8. In the R chunk below, copy and paste your code for your graph of price versus building_area, colored by has_fireplace from above. Then add a new layer `geom_line(mapping = aes(x = building_area , y = .fitted, color = has_fireplace))` to visualize the graph of the model.

```
ggplot(data = frederick_2019,
       mapping = aes(x = building_area,
                     y = price, color = has_fireplace)) +
  geom_jitter(shape = 1, size = 1, alpha = 0.5) +
  scale_y_continuous(labels = label_dollar()) + geom_line(mapping = aes(x = building_area , y = .fitted, color = has_fireplace))
```



As a class, we will talk about how to interpret the coefficients of our new model.

*If you are done before your classmates, please help us out by copying and pasting any new code from today's lab into our R cheat-sheet.